

Vrinda Store Sales Analysis

Project Overview

This portfolio showcases a comprehensive data analysis project focused on analyzing Vrinda Store's 2022 sales performance. The project demonstrates skills in data cleaning, exploratory data analysis, data visualization, and deriving actionable business insights.

Tools Used: Microsoft Excel, Pivot Tables, Data Visualization

PROJECT SUMMARY

This analysis examines Vrinda Store's 2022 sales data to identify key trends, customer behavior patterns, and growth opportunities for 2023.

Project Name	Vrinda Store Annual Sales Analysis 2022
Objective	Create an annual sales report to understand customer behavior and identify strategies to grow sales in 2023
Dataset Size	31,047 orders across 12 months
Date Range	January 2022 - December 2022
Tools Used	Microsoft Excel, Pivot Tables, Charts & Dashboards

Analysis Questions Addressed

- Compare sales and orders trends throughout the year
- Identify the month with highest sales and order volume
- Analyze purchasing behavior by gender (Men vs Women)
- Examine different order statuses and their distribution
- Identify top 10 states contributing to sales
- Understand relationship between age and gender based on order volume
- Determine which sales channels contribute maximum revenue
- Identify the highest selling product categories

DATA OVERVIEW

Dataset Structure

The dataset contains comprehensive e-commerce transaction data with 21 columns covering order details, customer demographics, product information, and shipping data.

Data Category	Fields
Order Information	Order ID, Date, Month, Status, Amount, Currency, Qty
Customer Data	Customer ID, Gender, Age, Age Group
Product Details	SKU, Category, Size
Sales Channel	Channel, B2B indicator
Shipping Info	Ship City, Ship State, Ship Postal Code, Ship Country

Key Statistics

Total Orders 31,047	Total Revenue ₹2.12 Crores
Average Order Value ₹682	37 States & 1861 Cities Covered

KEY FINDINGS

1. Monthly Sales Performance

The analysis reveals distinct seasonal patterns in sales throughout 2022:

- Peak Performance: March recorded the highest sales at ₹19.28 lakhs with 2,819 orders
- Strong Q1: First quarter (Jan-Mar) showed consistently high performance with sales above ₹18 lakhs
- Declining Trend: Sales gradually decreased from mid-year, with Q4 showing the lowest performance
- November recorded the lowest sales at ₹16.15 lakhs with 2,383 orders

2. Customer Demographics - Gender Analysis

Women constitute the primary customer base:

- Women: ₹1.36 crores (64% of total sales)
- Men: ₹76.14 lakhs (36% of total sales)
- Women customers contribute nearly 1.8x more revenue than men

3. Order Status Distribution

Order fulfillment shows strong operational efficiency:

- Delivered: 28,641 orders (92.3%) - Excellent fulfillment rate
- Returned: 1,045 orders (3.4%)
- Cancelled: 844 orders (2.7%)
- Refunded: 517 orders (1.7%)

The 92.3% delivery success rate indicates robust logistics and customer satisfaction.

4. Top Performing States

Sales concentration in major metropolitan regions:

State	Sales Amount (₹)
1. Maharashtra	29,90,221
2. Karnataka	26,46,358
3. Uttar Pradesh	21,04,659
4. Telangana	17,12,439

5. Tamil Nadu	16,78,877
---------------	-----------

These top 5 states collectively contribute over 55% of total revenue, with Maharashtra leading at ₹29.9 lakhs.

5. Age and Gender Distribution

Adult women represent the largest customer segment:

Age Group	Men %	Women %
Adult (30-50)	15.5%	34.6%
Teenage (18-29)	9.2%	21.1%
Senior (50+)	5.9%	13.7%

Key Insight: Adult women (30-50 age group) account for 34.6% of all orders, making them the most valuable demographic segment.

6. Sales Channel Performance

Amazon emerges as the dominant sales channel:

Channel	% of Orders
Amazon	35.5%
Myntra	23.4%
Flipkart	21.6%
Ajio	6.2%
Others (Nalli, Meesho, etc.)	13.3%

The top 3 marketplaces (Amazon, Myntra, Flipkart) account for 80.5% of all orders, indicating strong reliance on major e-commerce platforms.

INSIGHTS & RECOMMENDATIONS

Strategic Recommendations for 2023

1. Target the Primary Customer Segment

Focus marketing efforts on adult women (30-50 years) who contribute the highest order volume. Create targeted campaigns featuring products specifically designed for this demographic across top-performing states.

2. Strengthen Top Channel Partnerships

Invest in deeper partnerships with Amazon, Myntra, and Flipkart which collectively drive 80% of orders. Negotiate better placement, exclusive launches, and promotional opportunities on these platforms.

3. Geographic Expansion Strategy

While maintaining strong presence in Maharashtra, Karnataka, and Uttar Pradesh, explore growth opportunities in emerging markets. Consider regional marketing campaigns and improved logistics in tier-2 and tier-3 cities within these states.

4. Address Seasonal Decline

Sales showed a declining trend from Q2 onwards. Implement strategic promotions, festive campaigns, and new product launches during typically slower months (Sep-Dec) to maintain revenue consistency throughout the year.

5. Reduce Order Cancellations and Returns

While the 92% delivery rate is strong, the 7.8% combined cancellation and return rate represents potential revenue leakage. Improve product descriptions, sizing guides, and quality control to minimize returns and enhance customer satisfaction.

6. Expand Men's Category

Men currently represent only 36% of sales. This presents a significant growth opportunity. Develop and market men's product lines more aggressively to capture this underserved segment.

Expected Impact of Recommendations

Implementing these data-driven strategies could result in:

- 15-20% increase in overall sales for 2023
- Improved revenue stability across all quarters
- Higher customer lifetime value through reduced returns
- Expanded customer base through targeted demographic growth

TECHNICAL SKILLS DEMONSTRATED

Data Analysis Competencies

Skill Category	Skills Applied
Data Cleaning	Data validation, handling missing values, formatting consistency, data type standardization
Data Analysis	Pivot tables, aggregation functions, trend analysis, comparative analysis, segmentation
Visualization	Charts creation, dashboard design, data storytelling, visual best practices
Business Intelligence	KPI identification, trend interpretation, actionable insights generation, strategic recommendations
Tools	Microsoft Excel (Advanced), Pivot Tables, Charts, Formulas & Functions, Data Analysis Toolpak

Analysis Methodology

- **Data Collection:** Compiled 31,047 transaction records spanning entire year 2022
- **Data Preparation:** Cleaned and validated data, standardized formats, created calculated fields
- **Exploratory Analysis:** Used pivot tables to examine patterns across multiple dimensions
- **Visualization:** Created comprehensive dashboard with charts for easy interpretation
- **Insight Generation:** Derived actionable business recommendations from data patterns

CONCLUSION

This comprehensive analysis of Vrinda Store's 2022 sales data reveals clear patterns and opportunities for business growth. The data demonstrates strong performance with ₹2.12 crores in annual revenue across 31,047 orders, but also highlights areas for strategic improvement.

Key Takeaways

- **Primary Customer Base:** Adult women (30-50) represent the core demographic, contributing 34.6% of orders
 - **Channel Strategy:** Amazon, Myntra, and Flipkart drive 80% of business - deepening these partnerships is critical
 - **Geographic Focus:** Top 5 states generate 55% of revenue, with Maharashtra leading
 - **Seasonal Trends:** Q1 shows strong performance, with gradual decline through the year
 - **Growth Opportunity:** Men's segment is underdeveloped, presenting expansion potential
-